

ILES-PROJECT - Database Constraints

CONSTRAINTS

Concept	Meaning
unique	No duplicates
null=False	Must have value
default	Auto value
unique_together	Composite rule

Django model constraints are fundamental tools for **data integrity**, ensuring your database remains accurate and consistent even if bugs occur in your application code. While application-level validation (like forms) catches errors early for users, database-level constraints act as a final "safety net" that cannot be bypassed by manual scripts, the Django shell, or direct database access.

Core Constraints & Their Impact

unique=True

- **Function:** Ensures no two rows in a table have the same value for a specific field.
- **Integrity Role:** Prevents duplicate data (e.g., two users with the same email). It creates a **Unique Index** in the database, which also speeds up lookups.

null=False(Default):

- **Function:** Forces the database to reject any record where this field is empty (**NULL**).
- **Integrity Role:** Guarantees that essential data is always present.
- **Note:** For string fields (CharField, TextField), Django prefers storing an empty string (“ ”) rather than **NULL**, so null=True is typically avoided for strings unless combined with uniqueness.

default=:

- **Function:** Provides a fallback value if no data is supplied during record creation.

- **Integrity Role:** Prevents "breaking" the database when adding new required fields to existing tables. It ensures every record has a valid value, maintaining consistency across old and new data.

Unique_together (Legacy)/UniqueConstrain (Modern)

- **Function:** Ensures that a **combination** of fields is unique (e.g., a "Room" and "Date" combo for a booking).
- **Integrity Role:** Enforces complex business rules at the schema level.

Unique Constraints

- Email field in User model is unique
- EvaluationCriteria name is unique

Required Fields

- WeeklyLog content is required (null=False)

Default Values

- WeeklyLog status defaults to 'draft'
- EvaluationCriteria max_score defaults to 10

Composite Constraint

- A student cannot have multiple logs for the same week
- Implemented using unique_together (student, week_number)